DR. EWE WIN ENG

Data Scientist | Quantitative Researcher | Renewable Systems Specialist
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Visa Status:
UK Global Talent Visa Holder (Eligible to work immediately. No sponsorship required)

PROFESSIONAL SUMMARY

Ph.D.-qualified Data Scientist and Quantitative Researcher with expert proficiency in Python, TensorFlow, and Time-Series Forecasting. Specialises in bridging theoretical physics and commercial AI applications, with a proven track record of optimising complex energy systems and supporting high-stakes public sector initiatives. Combining statistical rigour with production-level engineering, seeking to drive commercial value in Glasgow's Data Science or FinTech sectors.

TECHNICAL SKILLS

- **Languages & Core:** Python (Expert: Pandas, NumPy, SciPy), SQL, MATLAB, Bash Scripting.
- **Machine Learning:** TensorFlow, Keras, PyTorch, Scikit-Learn, Deep Learning (ResNet-MLP), Genetic Algorithms.
- **Analytics & Maths:** Time-Series Forecasting, A/B Testing, Hypothesis Testing, Finite Volume Method, Regression Analysis.
- **DevOps & Tools:** Git/GitHub, Docker, Jupyter, Google Colab, Power BI, Data Visualisation (Seaborn/Matplotlib).

PROFESSIONAL EXPERIENCE

STEaM | Glasgow, UK | **Energy Data Scientist & Postdoctoral Researcher** | 2023 – 2025

- Engineered a commercial-grade 1D finite volume simulation engine in Python to model thermodynamic stratification, validating GigaWatt-hour thermal storage feasibility for industrial assets.
- Developed intelligent control algorithms for Heat Pump/CHP systems, linking grid balancing incentives directly to storage charging logic to maximise revenue capture.
- Conducted comprehensive techno-economic analysis (LCOH, COP) to de-risk the conversion of legacy mine shafts into renewable district heating assets.
- Built automated ETL pipelines using Pandas and SciPy to process high-resolution electricity demand profiles, enabling robust sensitivity analysis across varying tariff structures.

PVT SYSTEMS OPTIMISATION | Malaysia | **Simulation & Optimisation Specialist** | 2018 – 2022

- Designed and implemented custom Genetic Algorithms to iterate through thousands of design variables, simulating performance outcomes for solar collectors.
- Identified an optimal system configuration that increased energy capture efficiency by 30%, demonstrating direct ROI through algorithmic optimisation.

KEY TECHNICAL PROJECTS

Automated Valuation Engine (Diamond Price Prediction) | *Machine Learning Engineer*

- Engineered a production-ready valuation pipeline utilizing a custom ResNet-MLP architecture to predict asset prices with high precision ($R^2 > 0.95$).
- Implemented residual skip connections and Log-Norm target engineering to model complex non-linear interactions and stabilise gradients during training.

NHS Covid-19 Demand Modelling | *Data Scientist*

- Analyzed official NHS Scotland open data to model ICU bed usage during critical demand spikes.
- Developed a statistical framework for patient intake projection, demonstrating how data-driven insights support public health resource allocation.

EDUCATION

- **Ph.D. in Renewable Energy** | Universiti Kebangsaan Malaysia | 2018 – 2022
- **M.Sc. in Energy Technology** | Universiti Kebangsaan Malaysia | 2017 – 2018
- **B.Sc. in Physics** | University of Glasgow, UK | 2014 – 2017

ACHIEVEMENTS

- **Global Talent Visa:** Recognised by the UK government as a leader in digital technology and science.
- **Publications:** Author of multiple peer-reviewed papers on solar energy, renewable energy systems, and sustainable energy applications (https://scholar.google.com/citations?user=O1De6b8AAAAJ&hl=en)